

Linux File & Log Management in a Real-Time Production Environment

Learning Objectives:

By the end of this class, you will be able to:

- Perform real-time file and directory management tasks on a Linux-based server.
- Monitor and manage log files using standard Linux tools.
- Simulate a production scenario with system-generated logs.
- Apply troubleshooting using logs.

Real-Life Industry Scenario:

You are working as a **Cloud Support Engineer** at a SaaS company. Your responsibilities include managing user data stored in directories and troubleshooting application issues using Linux logs. Today, the DevOps team has reported performance issues, and the Support team is requesting data backup for a specific user.

Prerequisites:

- Running EC2 instance (Ubuntu)
- Create IAM user
 - Give EC2 full permissions
 - Access EC2 via IAM use

PART 1: Practical File Management Hands-On

Real-Life Scenario:

You are managing client data directories on a Linux server. Each client has their own folder with data and reports. Your job is to:

- Set up the structure
- Control access with permissions
- Archive reports for backup

✅ Step-by-Step Instructions with Real Output

🧱 Step 1: Create Client Directories

```
sudo mkdir -p /opt/clients/client_alpha/{data,reports}
sudo mkdir -p /opt/clients/client_beta/{data,reports}
sudo mkdir -p /opt/clients/client_gamma/{data,reports}
```

📦 Output:

(no output means success)

🔧 Step 2: Create User Groups

```
sudo groupadd client_alpha
sudo groupadd client_beta
sudo groupadd client_gamma
```

👥 Step 3: Set Ownership & Permissions

```
sudo chown -R root:client_alpha /opt/clients/client_alpha
sudo chmod -R 770 /opt/clients/client_alpha
```

```
sudo chown -R root:client_beta /opt/clients/client_beta
sudo chmod -R 770 /opt/clients/client_beta
```

```
sudo chown -R root:client_gamma /opt/clients/client_gamma
sudo chmod -R 770 /opt/clients/client_gamma
```

📁 Output (Check using `ls -ld`):

```
ls -ld /opt/clients/client_alpha
```

```
drwxrwx--- 3 root client_alpha 4096 May 12 10:01
/opt/clients/client_alpha
```

Step 4: Create Dummy Files in data/

```
echo "user_id,score" | sudo tee
/opt/clients/client_alpha/data/data1.csv > /dev/null
```

```
echo "1,98" >> /opt/clients/client_alpha/data/data1.csv
```

Fix: Use tee with sudo to append safely:

```
echo "1,98" | sudo tee -a
/opt/clients/client_alpha/data/data1.csv > /dev/null
```

```
=====
sudo touch /opt/clients/client_alpha/data/data2.csv
sudo touch /opt/clients/client_alpha/data/data3.csv
```

Fix: Allow directory traversal (execute permission) on parent directories

Run the following commands to allow traversal by the `ubuntu` user:

```
sudo chmod +x /opt
sudo chmod +x /opt/clients
sudo chmod +x /opt/clients/client_alpha
```

`+x` on a directory means users can "cd" into it or access files inside.

Output:

```
cat /opt/clients/client_alpha/data/data1.csv
```

```
user_id,score
1,98
```

Step 5: Archive All **reports/** Folders

```
sudo tar -czvf client_reports_backup.tar.gz -C /opt/clients  
client_alpha/reports client_beta/reports client_gamma/reports
```

Output:

```
client_alpha/reports/  
client_beta/reports/  
client_gamma/reports/
```

Step 6: Move and Set Permissions

```
sudo mv client_reports_backup.tar.gz /opt/backups/  
sudo chmod 444 /opt/backups/client_reports_backup.tar.gz
```

Final Check:

```
ls -l /opt/backups/client_reports_backup.tar.gz  
  
-r--r--r-- 1 root root 10240 May 12 10:10  
/opt/backups/client_reports_backup.tar.gz
```

PART 2: Practical Logs Management Hands-On

Real-Life Scenario:

Your server is under SSH brute-force attacks. You need to extract all SSH login attempts and summarize the log activity for analysis.

logrotate is a tool in Linux that helps manage your log files so they don't become too big and fill your disk.

What does it do?

It can:

- **Rotate** the log (rename it, start a new one)
- **Compress** old logs to save space
- **Delete** very old logs
- **Run a command** (like restarting a service) after rotating the logs

✅ Step-by-Step Instructions with Real Output

🗺️ Step 1: Navigate to Log Directory

```
cd /var/log
```

📁 Step 2: View System Log (Ubuntu)

```
sudo tail -n 10 /var/log/syslog
```

📄 Output Example:

```
May 12 10:12:01 ubuntu CRON[2101]: (root) CMD (test -x  
/etc/init.d/anacron && /etc/init.d/anacron start)  
May 12 10:12:01 ubuntu systemd[1]: Starting Clean php session  
files...
```

Creating Log Files

📁 Create Working Directory

```
mkdir -p /opt/log_reports  
cd /opt/log_reports
```



Create **auth.log** (dummy log with SSH entries)

nano auth.log (open auth.log file with nano editor)

Paste the following content:

```
May 25 10:00:01 server sshd[1234]: Accepted password for alice
from 192.168.0.2 port 54321 ssh2
May 25 10:00:02 server sshd[1234]: pam_unix(sshd:session):
session opened for user alice by (uid=0)
May 25 10:05:17 server sshd[1235]: Failed password for root from
192.168.0.3 port 54322 ssh2
May 25 10:07:45 server sshd[1236]: Accepted password for bob
from 192.168.0.4 port 54323 ssh2
May 25 10:07:46 server sshd[1236]: pam_unix(sshd:session):
session opened for user bob by (uid=0)
May 25 10:10:55 server sshd[1237]: Accepted password for charlie
from 192.168.0.5 port 54324 ssh2
May 25 10:10:56 server sshd[1237]: pam_unix(sshd:session):
session opened for user charlie by (uid=0)
```

Save and exit (CTRL+O, ENTER, CTRL+X).



Step 3: Filter SSH Logs

```
grep "sshd" /var/log/auth.log > /opt/log_reports/ssh_access.log
```



Output Example (inside file):

```
May 12 10:15:32 ubuntu sshd[2154]: Accepted password for
testuser from 192.168.1.15 port 53782 ssh2
May 12 10:15:33 ubuntu sshd[2154]: pam_unix(sshd:session):
session opened for user testuser by (uid=0)
```

Step 4: Extract Timestamp and Username

```
awk '/sshd.*Accepted/ {print $1, $2, $3, $9}'
/opt/log_reports/ssh_access.log > /opt/log_reports/summary.log
```

Output (inside summary.log):

```
May 12 10:15:32 testuser
```

Step 5: Archive Logs

```
tar -czvf /opt/log_reports/log_archive.tar.gz -C
/opt/log_reports ssh_access.log summary.log
```

Output:

```
ssh_access.log
summary.log
```

Step 6: Simulate Log Rotation

Create a logrotate config file

Create a file `/etc/logrotate.d/ssh_access`:

```
sudo nano /etc/logrotate.d/ssh_access
```

Paste these below lines into created file

```
/opt/log_reports/ssh_access.log {  
    size 10k  
    daily  
    rotate 4  
    compress  
    missingok  
    notifempty  
}
```

This Means:

- The log rotates when it reaches 10 KB.
- Keep **4 backups and delete old ones**
- **Compress** old logs
- **Do nothing** if log is missing or empty

Simulate logrotate:

```
sudo logrotate -f /etc/logrotate.d/ssh_access
```

This means:

- `sudo`: Run as admin
- `logrotate`: Use the logrotate tool
- `-f`: Force it to run **now**
- `/etc/logrotate.d/ssh_access`: Use the rules written in this file

Verify the rotation

Then check if the rotation happened:

```
ls -l /var/log/demo*
```


Summary / Conclusion

- As a Cloud Support Engineer, you manage client directories, secure user data, and perform backups for a SaaS company.
- Under a brute-force SSH attack, you analyze logs and summarize login attempts.

[Key Achievements]

- Secure file and directory access using groups and permissions.
- Backup and archive important data and logs.
- Use logs for troubleshooting and detect security issues.
- Automate log management with logrotate.

 Prepared you for real-world tasks as a Cloud Support or DevOps Engineer.